

Introduction:

Machine design: is defined as the use of scientific principles, technical information and imagination in the description of a machine or a mechanical system to perform specific function with maximum economy and efficiency.

Significance:

②

Kind of design work: There are two traditional methods in design.

①

Design by Evolution: → In the past the design to evolve over a long span of time. The rate of development was slow.

- Products remained static for quite some time. If changes or some improvements were made these were small and gradual.
- Each change only made a small improvement in a product which already existed.
- Most of the new designs were copies of existing design with minor changes to suit the needs of the local requirements.
- The circumstances rarely demanded the utmost skill and analytical capabilities of the designer.
- Such a situation is referred to as design by evolution in which the technical risk and stakes are proportionately low.

In design by evolution the designer has to weight a no. of conflicting requirements such as the following so that the ultimate product is reliable and satisfies the need of the customers.

- (a) Improvement in performance, quality and appearance.
- (b) Reduction in overall cost.
- (c) Changed market requirement and trend of competition.

- (4) Use of new or improved materials and manufacturing technologies.
- (5) Improved functionality of the product.

② Design by Innovation In today scenario design by evolution condition is not valid in many spheres of design activity.

→ Present circumstances require bolder and faster improvements. Consequently the present day designer faces a higher technical risk and has greater stakes.

→ Nowadays design are by innovation i.e. following any scientific discovery when a new product is developed, its proper use may dictate an almost complete break with past practice.

→ A new product is developed on ideas which were never practised or tried. As the risk of technical error is high, the skills and knowledge required have to be much greater. To help designers face this challenging environment various analytical tools such as computer-aided design, drafting, finite element method and simulation techniques have been developed to reduce the risk of technical error. However the job of the designers is still not fully simplified by these developments. They encounter a host of problems which are peculiar to the process of design. These are

① Need of developing, organizing and evaluating information almost always in the face of uncertainty.

(1)

(2)

Necessity of taking into account the complicated interaction of components.

(3)

Constant requirement to make predictions in terms of design criteria.

(4)

Need to work within the constraints of an economic framework.

Design Process: There are 3 phases which include the whole process of design.

(1) Conceptual design.

(2) Embodiment design.

(3) Detailed design.

(1) Conceptual design: In this phase problem is identified through abstraction. It involves establishment of structure, search for suitable solution principles and their combination into concept variants. The concept variants that have been elaborated have to be eliminated. The best must be judged by the application of specific criteria. The various steps of conceptual design are.

(1) To determine whether the need is original, whether it is valid and has current existence, or has strong evidence of latent existence.

(2) To explore the design problem engendered by the need and to identify its elements such as working parameters, constraints and major design criteria.

(3) To conceive a no. of plausible solution to the problem.

- (1) To shortlist the potentially useful solution of plausible ones on the basis of
- (a) technical suitability (b) physical realizability
 - (c) economic feasibility.

In conclusion the conceptual design indicates whether a current or potential need exists. What the design problem is; what are plausible solution of it? In other words the conceptual design explore the feasibility of the design project.

(2) Embodiment Design: This is the second phase of design process. In this phase preliminary design of the system starts with the set of useful solution which were evolved in the conceptual design phase. The purpose of this phase is to establish which of the proposed alternatives is the best design concept. The various steps involved in this phase of the design are.

- (1) To evaluate each available solution until one is proved to be the best.
- (2) To initiate synthesis studies for establishing the range within which the major design parameters must be controlled.
- (3) To investigate the tolerances in the characteristics of major components and critical materials that may fit properly into the system.
- (4) To examine the influence of environmental internal and external constituents on the system.

⑤ To study how the design will meet the consumers' need. How will it match with the product of competitors? whether scarce and critical raw materials will continue to be available? what would be the rate of obsolescence and its effect on product quality?

⑥ To test the critical aspect of the design in order to validate the design concept and to provide necessary information for its subsequent phases.

③ Detailed design: In this phase the engineering description of a tested and producible design is furnished. up to this pt. the design project is characterized by great fluidity. Major changes in concept could be accommodated without adverse financial loss. The various steps of engineering description of a feasible design are given below.

① Develop an overall synthesis of the design project and prepare a master layout of the system.

② Prepare specification of various sub-system and components on the basis of the master layout.

③ Decide various dimension of components and if required, make the necessary changes in the layout of sub-systems and in the master layout.

④ Initiate experimental design by constructing models to check untried ideas.

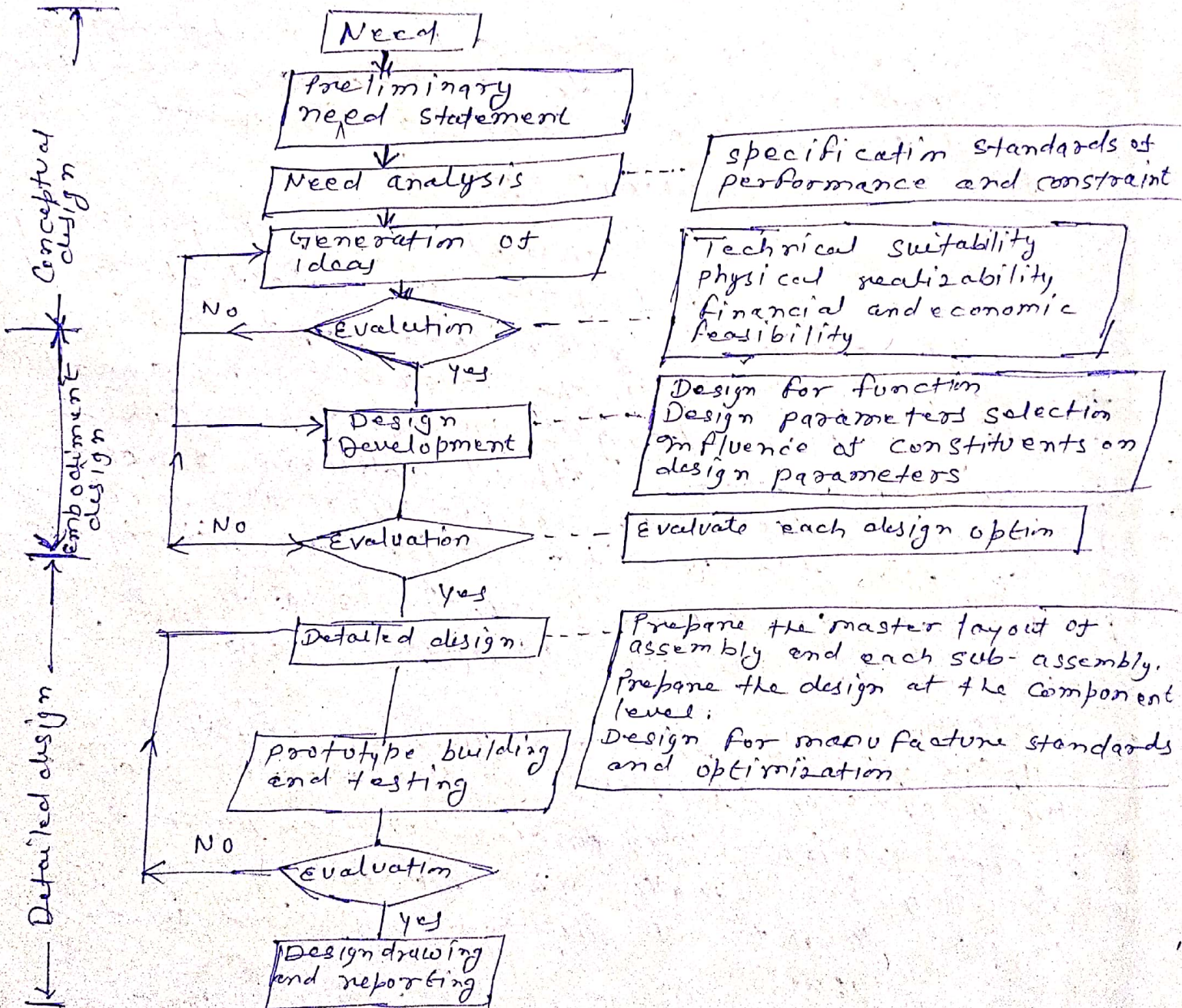
⑤ Test the prototype and obtain the necessary feedback, needed for redesign and improvement.

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Need analysis

In this the designer have to create the need of the product for ex. Mobile is the basic requirement now a days and the updation is available in the market every 6 month. That is the need which create by the designer.

Design through Morphological analysis



Designers responsibility introduction to various aspect of design

Manufacturing aspects of design categories into two.

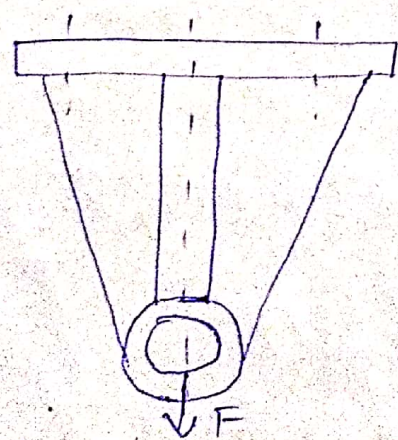
- ① Primary manufacturing method.
- ② Secondary ——— method.

- ① It consists of casting, welding, riveting and forging. In this raw materials is shape by heat.
- ② It consists required the exact dimensions. It consists turning, drilling, boring, milling, planing, shaping, broaching, grinding, honing and lapping.

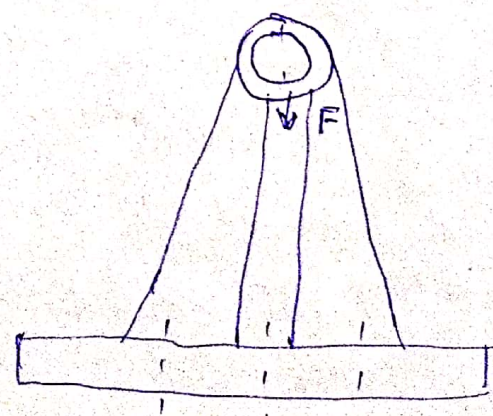
Design Consideration for casting: It is a basic manufacturing process in which the metal is first liquified and after properly heating it in a suitable furnace. The liquid or molten metal is poured into a previously prepared mould cavity where it is allowed to solidify.

The designer should follow the following recommendations to minimize the casting-related problem.

- ① Cast iron has more strength in compression than in tension.



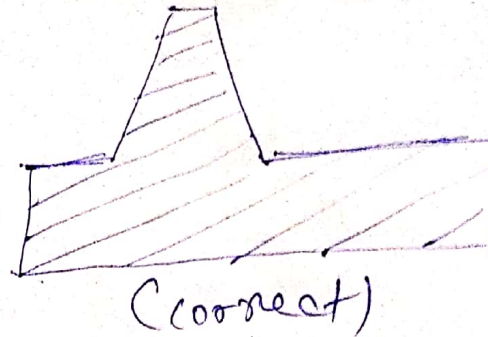
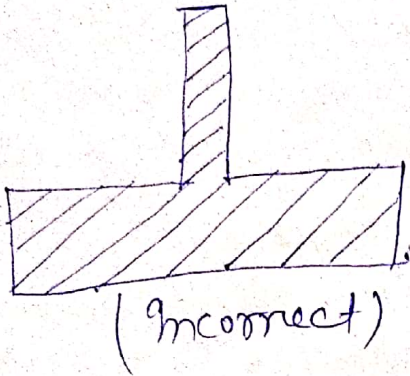
① Incorrect (Tension)



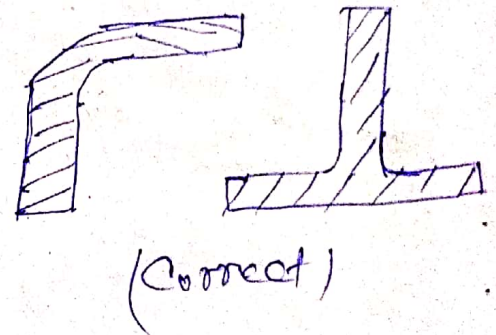
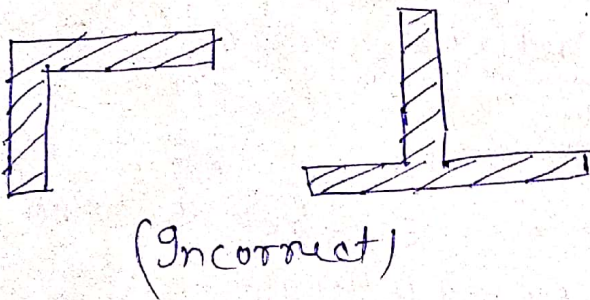
② Correct (Compression)

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- ② The section thickness should preferably be uniform and compatible with the overall design consideration. Any abrupt change in cross-section should be avoided.



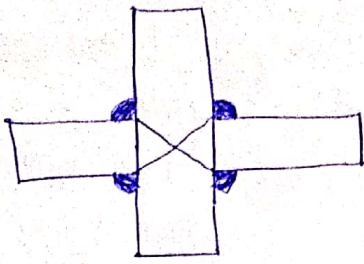
- ③ Avoid Concentration of metal at the junction and round all external corners by suitable fillet radii.



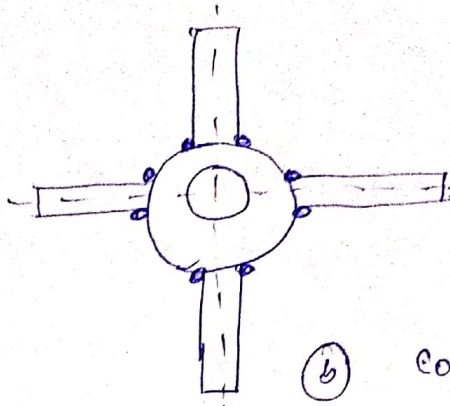
Design Consideration for welded parts:-

These are the following aspects which we should follow.

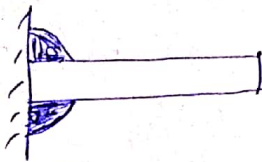
- ① Concentration at welds should be avoided at any one place.
- ② Avoid welding at inaccessible place.
- ③ Preferably use butt welding and avoid lap, stoop and stiffenings angles.
- ④ Control the profile of the weld. Avoid convex profile as it may cause stress concentration.



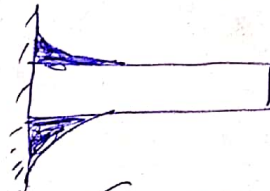
(a) Incorrect weld pt.



(b) correct weld pt.

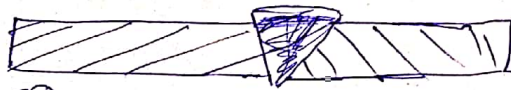


(a) (Incorrect)



(b) (Correct)

⑤ In order to avoid stress concentration; a double V-butt joint is preferred to a single V-butt weld.



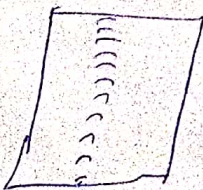
(a) Single V-butt weld.



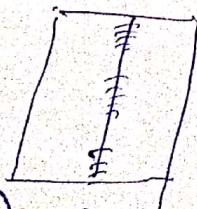
(b) Double V-butt weld.

⑥ Do not weld steel of high Hardenability it may result into cracking.

⑦ If possible intermittent welding should be used, it reduce the possibility of distortion.



(a) Long Continuous weld

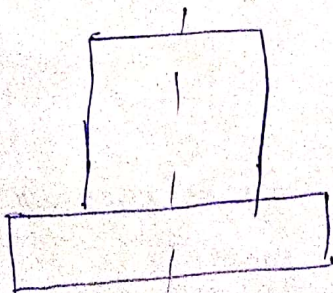
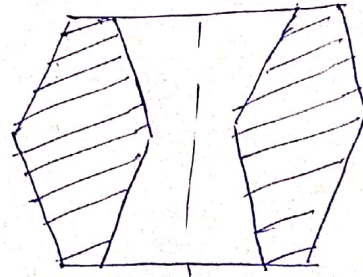
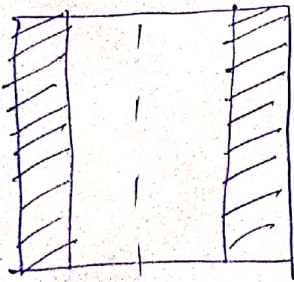


(b) Intermittent weld.

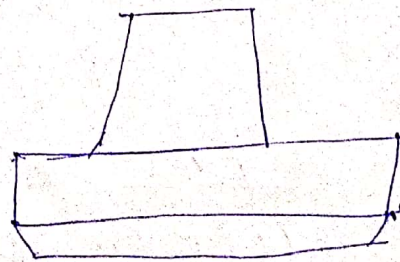
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Design Consideration for forgings. There are the following aspects to consider while designing.

- ① The part should be shaped in such a way that the fibre lines remain parallel to the tensile force and perpendicular to the shear force.
- ② In the forged part, the deep cuts of machining should be avoided to prevent the fibre line from being broken. Broken fibre lines result into a weak part.
- ③ In drop forging - a die should be provided with a sufficient draft angle, usually $7-15^\circ$.



Original product



Modified product

Design Consideration for Machining:

- ① As far as possible, the geometrical configuration of the machined surface should be

plane or circular because ~~at~~ any other type of surface would require the use of special tools and fixtures.

- ② Adequate tolerance should be prescribed for various surfaces. Tight tolerance would increase the cost of production.
- ③ A sufficiently large gripping surface should be provided on parts for machining surface.
- ④ The design should involve minimum machining to reduce the cost.
- ⑤ Whenever possible, the sharp corners and shoulders should be avoided. Adequate chamfer or fillet radii should be provided.